Supplement: Verifier-Guided Selective Repair for Diffusion-Generated Macro Placements

Anonymous Authors

April 26, 2026

A Implementation details of the verifier

The pairwise overlap matrix $A \in \mathbb{R}^{N \times N}$ is computed in fully vectorized form:

```
# centers: (N, 2), sizes: (N, 2)
cx, cy = centers[:, 0], centers[:, 1]
sw, sh = sizes[:, 0], sizes[:, 1]
ovx = torch.clamp_min(
    torch.minimum(cx[:,None]+sw[:,None]/2, cx[None,:]+sw[None,:]/2)
  - torch.maximum(cx[:,None]-sw[:,None]/2, cx[None,:]-sw[None,:]/2), 0)
ovy = torch.clamp_min(
    torch.minimum(cy[:,None]+sh[:,None]/2, cy[None,:]+sh[None,:]/2)
  - torch.maximum(cy[:,None]-sh[:,None]/2, cy[None,:]-sh[None,:]/2), 0)
A = ovx * ovy
A.fill_diagonal_(0)
```

Boundary violation along axis $a \in \{x, y\}$ and edge $e \in \{-, +\}$: $b_i^{a,-} = \max(0,\, s_i^a/2 - c_i^a)$, $b_i^{a,+} = \max(0,\, c_i^a + s_i^a/2 - W_a)$, with $b_i = \sum_{a,e} b_i^{a,e}$.

B Lambda sweep: per-circuit detail

The cross-circuit summary appears in the main paper. Table 1 gives the per-circuit, per-λ Pareto numbers behind that summary ($n = 3$ seeds per cell, mean over seeds). Note the strict-Pareto rate per operating point: from 0/18 at $\lambda = 0$ (pure repulsion, HPWL inflates $\sim 162\%$) to 16/18 at $\lambda = 4$.

	λ=0		λ=0.25		λ=0.5		λ=1		λ=2		λ=4	
Circuit	Δv	Δh	Δv	Δh	Δv	Δh	Δv	Δh	Δv	Δh	Δv	Δh
adaptec1	−41.7	+226.2	−39.5	+87.1	−40.1	+47.5	−36.4	+31.1	−37.0	−7.3	−34.5	−3.7
adaptec2	−55.5	+251.0	−48.2	+128.1	−45.8	+85.4	−44.6	+50.3	−46.6	+16.7	−39.9	−19.1
adaptec3	−51.0	+124.0	−50.6	+75.9	−50.9	+56.4	−51.0	+34.5	−52.9	+17.5	−55.2	+3.4
adaptec4	−62.1	+104.3	−60.9	+62.7	−60.6	+36.3	−60.8	+6.9	−58.4	−11.4	−56.8	−25.5
bigblue1	−20.0	+142.1	−18.7	−22.3	−22.9	−61.2	−23.4	−77.0	−25.5	−73.5	−22.6	−75.9
bigblue3	−78.6	+122.0	−78.4	+72.3	−79.0	+54.5	−79.2	+37.4	−79.3	+19.5	−78.5	−3.9
strict-P	0/18		2/18		3/18		4/18		8/18		**16/18**	

Table 1: Per-circuit $\Delta v\,\%$ and $\Delta h\,\%$ across the λ sweep, mean over 3 seeds. Strict-Pareto count rises monotonically with λ, peaking at 89% (16/18) at λ=4.

C Component ablation: full table

The main-paper component ablation runs each of the 8 variants on 6 circuits $\times$ 3 seeds = 18 paired trials, replicating the same ChipDiffusion-guided draft for each variant. Table 2 reproduces it verbatim. Two notes: (i) "− overlap signal" and "− repulsive force" produce identical numbers because in our implementation the repulsive force is the only consumer of the pairwise overlap matrix (this is by design and stated in §3.1 of the main paper); (ii) the "no boundary signal" variant is incidentally redundant with repulsive+attractive forces on these drafts because every macro overlaps a neighbor and the repulsive force pushes incidentally back inside the canvas.

variant	$\Delta v\,\%$	$\Delta h\,\%$	strict-Pareto	time (s)
full (control)	$-50.1_{\pm 17.1}$	$-5.1_{\pm 34.8}$	9/18	1.89
severity soft-mask	$-21.8_{\pm 16.9}$	$+43.7_{\pm 23.5}$	1/18	1.55
− overlap signal	$+79.2_{\pm 57.4}$	$-85.0_{\pm 7.2}$	0/18	1.43
− boundary signal	$-82.9_{\pm 4.1}$	$-8.9_{\pm 30.8}$	9/18	1.72
− attractive force	$-51.4_{\pm 18.5}$	$+138.4_{\pm 53.6}$	0/18	1.44
− repulsive force	$+79.2_{\pm 57.4}$	$-85.0_{\pm 7.2}$	0/18	1.46
random selector	$-50.1_{\pm 17.1}$	$-5.1_{\pm 34.8}$	9/18	1.58
uniform (all macros)	$-50.1_{\pm 17.1}$	$-5.1_{\pm 34.8}$	9/18	1.49

Table 2: Component ablation, mean ± std over 18 (circuit, seed) pairs. Same as main paper Table 6. Raw data: `results/vsr_extra/component_ablation.json`.

D Failure-mode analysis: full list

Table 3 lists every (circuit, seed) trial flagged as not strict-Pareto improving the raw draft at $\lambda = 2$. Of the 12 non-Pareto trials, all share the structural pattern described in §4.6 of the main paper: heavy-overlap drafts where repulsive force dominates attractive force, particularly on circuits with hub macros (adaptec3, bigblue3) where moving a single macro drags long edge tails. $\lambda = 4$ flips most of these into strict-Pareto territory at the cost of ~ 2 pp violation reduction.

Circuit	seed	$\Delta v\,\%$	$\Delta h\,\%$	failure flags
adaptec1	123	−37.5	+42.2	HPWL> +20%, non-Pareto
adaptec2	123	−60.5	+6.4	non-Pareto
adaptec2	300	−48.2	+25.7	HPWL> +20%, non-Pareto
adaptec2	2024	−61.8	+5.0	non-Pareto
adaptec3	42	−57.9	+23.1	HPWL> +20%, non-Pareto
adaptec3	123	−52.9	+26.9	HPWL> +20%, non-Pareto
adaptec3	300	−51.8	+33.9	HPWL> +20%, non-Pareto
adaptec3	2024	−51.2	+35.3	HPWL> +20%, non-Pareto
bigblue1	123	−13.9	−80.8	weak-violation-cut
bigblue1	2024	−20.0	−47.8	weak-violation-cut
bigblue3	42	−77.7	+15.0	non-Pareto
bigblue3	123	−81.5	+17.8	non-Pareto
bigblue3	300	−72.8	+14.2	non-Pareto
bigblue3	2024	−77.1	+12.9	non-Pareto

Table 3: All 14 trials flagged at λ=2. "HPWL> +20%" indicates HPWL inflation greater than 20 pp; "non-Pareto" indicates $\Delta v < 0$ but $\Delta h \geq 0$; "weak-violation-cut" indicates $\Delta v > -20\%$. Two bigblue1 trials are flagged purely on weak violation reduction (HPWL is actually strongly improved). Raw data: `results/vsr_extra/failure_cases.csv`.

E Toy cross-domain experiment

We trained a tiny ($\sim$ 200 K-parameter) attention-GNN-style denoiser on 1000 synthetic legal 30-disk layouts (canvas $[0,1]^2$, radii $\in [0.018, 0.035]$, sparse random graph with 60 edges) for 30,000 training steps. At inference, we sample 50 test prompts $\times$ 5 seeds = 250 runs per method. Numbers in Table 4 (mean over 250 runs).

Method	violation count	overlap area	wirelength
baseline (raw)	11.2	0.246	42.4
classifier guidance	27.8	**0.000**	227.9 (5.4× baseline)
RePaint, binary mask	**3.4**	0.064	38.6
VSR-post (ours)	6.2	**0.030**	41.0
VSR-intra (soft)	8.3	0.149	42.3

Table 4: Toy 2D non-overlap, mean over 250 runs/method. Two observations: (i) classifier guidance with strong constraint weighting kills overlap but inflates wirelength 5.4×, mirroring the placement-domain finding; (ii) RePaint with binary mask minimizes pair count but produces residual small overlaps; VSR-post minimizes the total overlap area while preserving wirelength. Raw rows: `results/toy/toy_results.json`.

The cross-domain pattern is consistent with the placement results: classifier guidance trades constraint satisfaction for wirelength inflation; verifier-guided structured repair preserves both. We treat this as a controlled domain-independence test, not a thorough cross-domain study.

F Neural repair operator: full training details

The negative-result GNN repair operator referenced in §5 of the main paper spans six training regimes:

Architecture. 3-layer GATv2 (heads= 4, hidden= 128, $\sim$ 100 K parameters), scale-invariant input normalization (centers and severity divided by $\max(W, H)$). Node features: [severity, boundary, n_overlappers, s^w, s^h]. Edge features: 4-dim pin offsets from ChipDiffusion's `edge_attr`.

Loss. L2 on the predicted displacement against a target defined by the variant.

Variants.

- **Synthetic teacher**: hand-crafted operator outputs on synthetic graphs (2500 train, 200 val).
- **Real teacher**: hand-crafted on diffusion-sampled ISPD2005 placements (3 samples $\times$ up to 3 circuits = 3–9 samples).
- **Data augmentation**: per-sample $\times$50 random perturbations.
- **Trajectory distillation**: (x_k, x_{k+1}) pairs from every hand-crafted step; 30 steps $\times$ 3 samples = 90 pairs.
- **Ground-truth target**: ISPD2005 industrial `.pl` placements.
- **Residual learning**: predict $x_{GT} - x_{\text{hand}}$ after hand-crafted convergence.

Findings. Best validation L2 was 0.0037 (trajectory distillation), but at inference on ISPD2005 the best variant (residual s=0.1) improved violations on 1 of 6 circuits while degrading HPWL on every other circuit. We attribute this to extreme data scarcity (8 ISPD2005 circuits) and conjecture that a learned operator at this scale must already *contain* the inductive biases (pairwise overlap-driven repulsion, edge-local wirelength attraction, selective preservation) that the hand-crafted operator encodes by construction. Detailed per-circuit evaluations: `results/neural_vsr/eval_*.json`.

G Convergence curves

Figure 1 shows per-iteration (v_t, h_t) trajectories for $T = 300$ iterations on adaptec1 and bigblue1 at $\lambda = 2$. Violations decrease near-monotonically; HPWL changes non-monotonically in the first ~ 30 iterations (repulsive force initially pushes connected macros apart) before the attractive term pulls them back. Plateau is reached by $T \approx 100$, confirming our default budget.

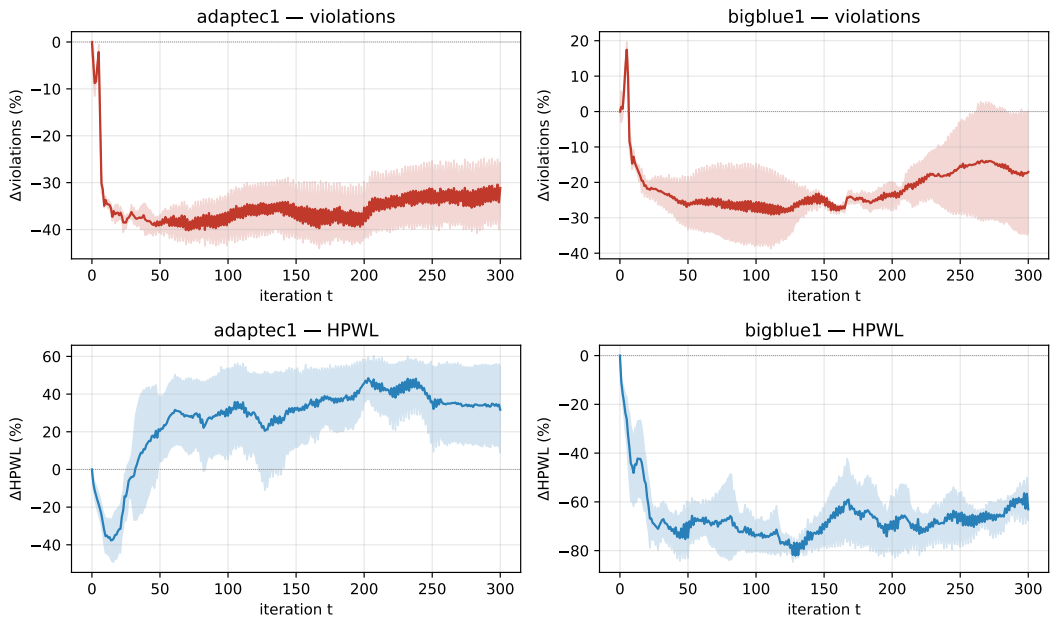

Figure 1: Per-iteration convergence on adaptec1 and bigblue1, 3 seeds. Top row: violations. Bottom row: HPWL. Shaded band: seed std.

H Memory profile of the diffusion backbone

In addition to the consolidated table in the main paper, the per-precision profile reveals two facts about the OOM frontier:

- Forward-step memory of the attention-GNN scales roughly linearly with N (macros): 221 MB on adaptec2 (N=566), 1030 MB on bigblue4 (N=8170), 2946 MB on bigblue2 (N=23084). This suggests the attention key/query/value tensors, not message passing, are the bottleneck.
- Full-rollout memory (100 denoising steps) saturates near 78 GB on the OOM circuits, consistent with KV-cache-like activation accumulation across the rollout that flash-attention does not fully release.

A natural future direction: replace the backbone's quadratic attention with linear attention (Performer-style kernelized attention) on the macro-graph axis; this would unblock bigblue2/4 evaluation but requires backbone retraining, which is beyond our scope.

Raw rows: `results/ispd2005/mem_profile.json` (9 rows: 3 circuits $\times$ 3 dtypes).

I Step-budget and step-size figures

Main paper Fig. 3 shows the cross-circuit step budget. Figs. 2 and 3 show the step-size and original step-budget ablation; they confirm T=100 and η=0.3 as robust defaults.

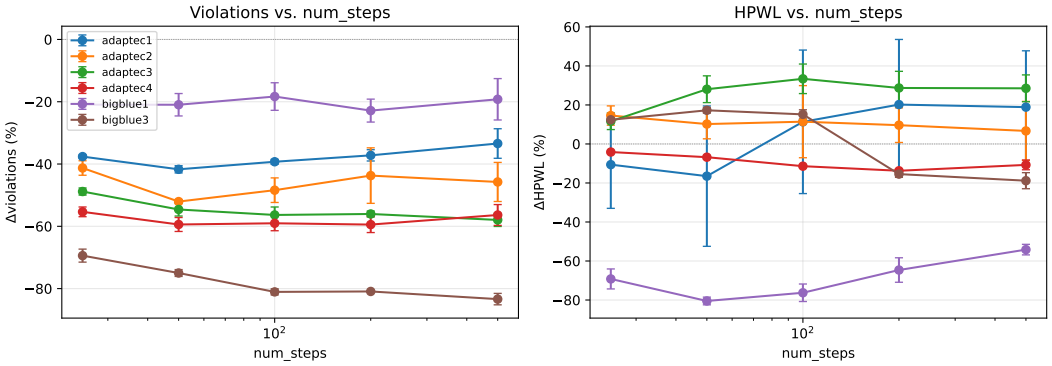

Figure 2: Number of repair steps ablation.

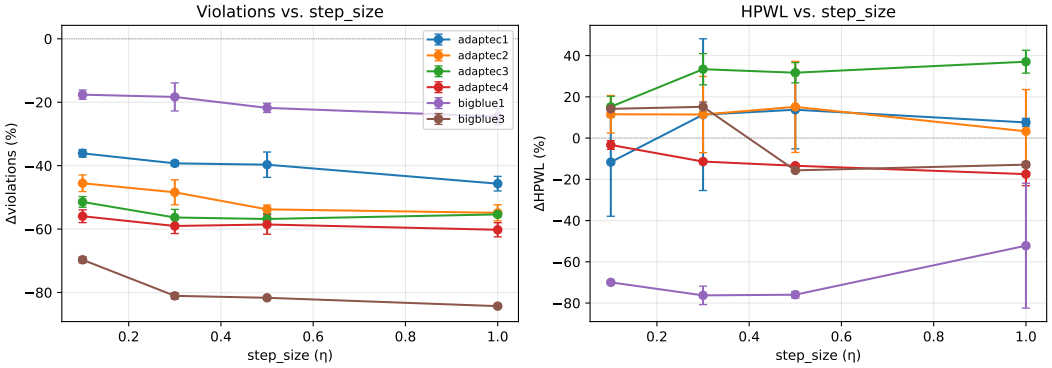

Figure 3: Step size ablation.

J Reproducibility and released artifacts

All code, configs, and result JSONs are released under the supplementary archive. The main-paper numbers reproduce from the following commands on a single A800 80 GB GPU (full sweep ~ 70 min):

```
python scripts/run_main_neurips.py        # main result, ~30 min
python scripts/run_timestep_sweep.py      # timestep sweep, ~12 min
python scripts/run_mem_profile.py         # memory profile, ~16 min
python scripts/run_component_ablation.py  # component ablation, ~12 min
python scripts/compute_neurips_stats.py   # main + supplement tables
python scripts/analyze_extra_experiments.py  # lambda/step/failure
python scripts/analyze_component_ablation.py # supplement Table 3
```

Seeds are fixed ({42, 123, 300, 2024} for the 4-seed runs; {42, 123, 300} for 3-seed sweeps). All result JSONs are under `results/ispd2005/` and `results/vsr_extra/`.